

A novel solar heating building integrated heat pipes and PCMs: Optimizing thermophysical properties and reducing energy consumption

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ABSTRACT

A novel building integrated heat pipes and phase change materials (BIHP-PCM) for solar heating is proposed. Using the variable thermal conductivity of heat pipes and the variable specific heat of phase change materials (PCM), BIHP-PCM greatly improves the indoor thermal environment in winter and reduces heating energy consumption. The heating performance of BIHP-PCM is closely related to the thermophysical properties of the PCM of the interior wall. In this study, volumetric phase change enthalpy, phase change temperature, and thermal conductivity of PCM in BIHP-PCM with and without HVAC are optimized at 61 cities in the severe cold and cold regions of China. It is found that the optimal PCM should be of high thermal conductivity and maximum phase change enthalpy in the given ranges. For phase change temperature, the optimal results with HVAC are slightly higher than the lower temperature limit of thermal comfort zone T_L by about 2 °C at all cities, and those without HVAC are lower than T_L at some cities because of the bottom-supporting effect on room temperature. The optimal energy saving ratio and improvement of thermal discomfort of BIHP-PCM at all cities are between 30 % and 100 % and are linearly correlated with the ratio of radiation to temperature difference of the local heating season. The application potentiality of BIHP-PCM is categorized into three zones, namely zero-carbon, good, and suitable. This study is helpful for promoting the application of BIHP-PCM to utilize solar energy effectively and efficiently to reach clean heating.

1. Introduction

China has put forward the strategic goal of peaking carbon emissions by 2030 and achieving carbon neutrality by 2060, and promoting energy efficiency in buildings is an important measure to achieve this dual-carbon goal. In China, heating energy consumption accounts for 27.0 % of building operation energy consumption in urban areas and 41.5 % of that in rural areas [1]. China is rich in solar energy resources [2], and solar buildings utilize solar energy for indoor heating, which can effectively decrease heating energy consumption and thus reduce the dependence on fossil fuels. Making full use of solar heating technology to meet the thermal demand of the indoor environment is an effective means of clean heating and plays an important role in promoting the realization of zero-carbon heating.

However, solar energy is a variable and intermittent resource [3].

When solar radiation is very weak during the daytime and not available at night, the intensity of the directly usable heat source is significantly insufficient, and how to ensure indoor thermal comfort in this circumstance is an inevitable challenge for the widespread application of solar buildings. At present, a popular solution is to integrate the phase change material (PCM) into the building envelope, such as walls, floors, and roofs, to improve the heat storage capacity of the building [4]. Compared with conventional building materials, PCM takes advantage of the latent heat of the phase change process rather than sensible heat for heat storage and release and has a large heat storage capacity in a narrow temperature range [5]. Buildings integrated with PCM store heat when solar radiation is abundant and release heat indoors when solar radiation is insufficient, to achieve indoor thermal comfort and save heating energy consumption [6].

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1.1. PCM in solar heating buildings

Many solar heating buildings integrated with PCM have been proposed in recent years and can be divided into passive and active applications according to whether the heating system leads to additional energy consumption [7]. Many studies concentrate on the three common passive solar house designs: the direct-gain house, the Trombe wall, and the attached sunspace. Guo et al. [8] applied PCM to the inner surface of the building envelope to form a triple-layer wall, and PCM received solar radiation through the window to absorb heat and released heat indoors by natural convection. The daily indoor temperature fluctuation was reduced, but the average room temperature did not change. Zhu et al. [9] analyzed the thermal performance of a Trombe wall with a double-layer PCM panel, which absorbed and released heat carried by hot air through natural convection, and the results showed the average indoor temperature was only 0.11 °C higher than a reference room. Vukadinović et al. [10] analyzed residential buildings with a sunspace and a thermal storage wall made of concrete containing PCM in Serbia, and the results indicated that the use of PCM saved only 1 %–3 % of the energy consumption for heating and cooling. It can be seen that for various passive building envelopes with PCM, the room temperature increase is limited, and the heating effect is not satisfactory. The reason is that the heat transfer coefficient of natural convection is low, and it is difficult to store solar heat and release heat to the indoor environment by PCM effectively and efficiently.

The active application of PCM in solar heating buildings mainly refers to the heating system with air or water as the heat transfer medium and with a supporting power and pipeline system. Qiao et al. [11] designed a solar wall combined with PCM and water capillary pipes, which required additional outdoor solar collectors and a water pump to transport hot water to release heat indoors. The average room temperature reached 25.5 °C and the indoor surface temperature was kept within the comfort range for more than 70 % of the test day. Zhou et al. [12] studied a ventilated Trombe wall with double PCM wallboards, using fans to form forced convection to enhance the heat transfer between air and PCM, and contributed to a reduction of 12.7 % in the heating load. Plytaria et al. [13] evaluated energetically a solar-assisted heat pump underfloor heating system with PCM, and the heating load was reduced by 40 %. It can be seen that compared with passive applications, the heating effect and the indoor thermal environment of active applications are improved. One of the reasons is that active applications increase the heat exchange intensity between the medium and the PCM by forced convection rather than natural convection so that the PCM can store and release heat more efficiently. However, these systems are often complex and cause additional energy consumption, with problems such as medium leakage and liquid frozen, resulting in high maintenance costs.

Gong et al. [14] proposed a solar heating building integrated with L-shaped flat gravity heat pipes (BIHP) recently. The L-shaped flat gravity heat pipe (HP) has the thermal diode characteristic and ultra-high thermal conductivity [15]. During the day, the HP effectively transfers the solar heat absorbed at the outer surface of the south wall to the east and west interior walls, and the interior walls in contact with the HP absorb heat through heat conduction. The ultra-high heat transfer of the HP is triggered automatically due to the temperature difference between the evaporation and condensation sections and requires no additional energy consumption by the gravity backflow of working fluid. At night, the interior walls release the stored heat into the room to maintain indoor thermal comfort. The BIHP can efficiently transfer solar heat to the room and improve the effective use of solar energy [16].

However, the above research about BIHP only focuses on interior walls constructed with conventional building materials with constant thermophysical properties. The heat transferred through the HP cannot be largely stored by conventional building materials, which leads to excessively high room temperature during the daytime and a rapid room temperature drop at night. Compared with conventional building

materials, PCM has the advantage of large heat storage capacity in a small temperature range, which can better store solar radiation heat. Therefore, if the interior wall of BIHP is constructed with PCM, PCM can be directly in contact with HP and store a large amount of heat through heat conduction, which can greatly improve the current inefficient utilization of PCM due to the low heat transfer coefficient of heat convection, without additional energy consumption required by fans and pumps.

1.2. Optimization of PCM for solar heating

As the main material to store and release heat in solar heating buildings, PCM affects the indoor heating significantly. Selecting appropriate phase change parameters for different buildings in different climate zones and different seasons is crucial [17]. Different from conventional building materials, the design of PCM needs to consider its unique thermophysical properties, such as phase change temperature, latent heat, thermal conductivity, phase change temperature range, thickness, and position [18].

For the phase change temperature, an excessively high value renders the melting condition difficult to achieve, while a value that is too low compromises indoor thermal comfort maintenance. Some studies indicate that the appropriate phase change temperature of PCM used for heating is generally several degrees higher than the indoor air temperature to effectively release heat to the indoor environment [19,20]. For the latent heat, the PCM with a larger latent heat can absorb and release more heat during the phase change process, but some research points out that it should be in a reasonable range instead of being too large or too small [21,22]. The thermal conductivity affects the heat transfer rate during the phase change process and a high thermal conductivity is conducive to the rapid storage and release of heat by PCM, but the marginal benefit of increasing the thermal conductivity will be significantly weakened with already a high value [23]. For the phase change temperature range, a narrow phase change temperature range means that PCM can store and release a large amount of heat with a small temperature fluctuation, and the regulation of room temperature is consequently more obvious [24]. For thickness, the amount of PCM needs to provide sufficient heat storage capacity to stabilize room temperature, and the optimal thickness varies according to its thermal properties [25]. For position, the optimal position of PCM is influenced by the composition and orientation of the phase change wall [26]. In general, the optimization of the above thermophysical properties requires comprehensive consideration of multiple factors.

It is crucial to choose an appropriate method in the optimization of phase change buildings for solar heating. Presently, there are mainly two optimization methods, i.e., based on direct problem idea and inverse problem idea [27]. The direct problem-based optimization method refers to selecting the optimal parameters from several simulated working conditions under the given thermophysical properties of the building envelope [28]. However, there are usually multiple factors, and the combination of the optimal solutions of each factor is usually not the global optimum. The inverse problem-based optimization method is a method that uses reverse thinking to find an accurate design solution as the optimal solution, which effectively avoids the step of trial and error during the solution process [29]. Zeng et al. [30,31] proposed the sequential quadratic programming method for building a sustainable building energy storage envelope, and the results showed that the theoretically optimal form of the specific heat of the envelope is a δ function, which is consistent with the thermophysical property of PCM. Particle swarm optimization is another commonly used inverse problem-based method for optimizing PCM in building energy performance. Soares et al. [32] used particle swarm optimization to evaluate the impact of PCM-drywalls in the energy savings of a lightweight building, including enthalpy, thermal conductivity, and thickness of PCM. Particle swarm optimization was also employed to optimize phase change heat storage and heat exchange systems applied in buildings [33,

34]. Particle swarm optimization is a well-developed technique to deal with nonlinear and nonconvex constrained problems, and uses the information shared by individuals inside the group to form an ordered movement of the whole particle swarm to obtain the optimal solution [35].

The interior envelope is the core heat storage body and heat release terminal of BIHP, and adjusting the thermophysical properties can optimize its heat storage and release capacity to further improve the indoor thermal environment. Kou et al. [36] proposed an efficient method based on particle swarm optimization to optimize the thermophysical properties of the BIHP interior envelope, and the optimal application effect in improving indoor thermal discomfort of BIHP made of conventional building materials without heating, ventilation, and air conditioning equipment (HVAC) in northern China was calculated. Thermal conductivity and volumetric specific heat capacity of the wall material, and the embedding depth ratio of the heat pipe were the optimization parameters. The above research only focuses on conventional building materials and does not consider PCM. Besides, previous research results show that although BIHP can greatly improve the indoor thermal environment, it is still possible that the natural room temperature of BIHP without HVAC drops lower than the lower limit of the thermal comfort temperature zone. In fact, buildings are usually equipped with HVAC to guarantee indoor thermal comfort, at the cost of heating energy consumption caused by HVAC. Therefore, the evaluation indexes of the indoor thermal environment of a room with or without HVAC are completely different. BIHP equipped with HVAC is more common in practical applications, however, research is still lacking on its optimal design and energy saving effect.

1.3. Climatic resources evaluation

The optimal heating performance of solar heating buildings with or without HVAC are various at different cities. Climatic condition varies greatly among regions, and the heating potential is affected by the heat supply and demand determined by the local climatic resources. Therefore, it is important for the application of solar heating buildings to select an appropriate climatic resource evaluation index and find the relationship between the local climatic resources and the optimal local heating performance. Thus, the solar heating potential can be pre-evaluated based on the local climatic resources without complex numerical simulation at the case level.

For a solar heating building integrated with PCM, climatic resources mainly refer to local solar radiation and outdoor air temperature. Many evaluation indicators have been proposed from the perspectives of total amount, intensity, and duration. Xiong et al. [37] proposed a two-tier method of climate zoning for climate-responsive building design, based on first indices including heating degree day, and second indices including mean daily total solar radiation in winter. Oliveira et al. [38] used the comfort solar fraction, which is a function of the solar/load ratio, the building thermal inertia, and the monthly non-utilizability, as an index to evaluate the heating effect of a direct-gain house. Fawaiier et al. [39] evaluated the influence of average solar radiation and heating degree day on the performance of the transpired solar collector. A Chinese technical code [40] uses the ratio of solar radiation to air temperature difference of the coldest month, which is defined as the ratio of the average solar radiation intensity to the temperature difference between indoor and outdoor air temperatures. This index takes the coldest month as the calculation period to characterize the most unfavorable condition for solar heating, but the local heating seasons at different cities are different, including the start date and time span. Therefore, this index cannot pertinently represent the local climatic resources during the actual period when indoor heating is needed.

Gong et al. [14] simulated the heating effect of BIHP made of conventional building materials in Beijing in January, but the conclusions were limited since only a single city and a single period were taken into consideration. Kou et al. [41] pointed out that compared with climatic

resource evaluation indexes such as heating degree day and average solar radiation intensity, the ratio of solar radiation to air temperature difference of the local heating season better reflected the improvement degree of indoor thermal discomfort of BIHP. However, the above research is limited to the BIHP of conventional building materials without HVAC, and the reduction of heating energy consumption of BIHP with HVAC has not been investigated. In addition, the coupling relationship between the heating performance of the BIHP integrated with PCM and local climatic resources is a research gap.

1.4. Aim of the present study

From the above literature review, it can be seen that the traditional passive applications of PCM in solar heating buildings are limited on the heating effect due to the heat transfer form, which is natural convection, while traditional active applications require additional systems and cause energy consumption. Combining PCM with heat pipes can effectively improve the heat transfer in absorbing and releasing solar heat by heat conduction without energy consumption. Because of the change in the heat transfer form of PCM, the thermal optimization and the best heating effect of this novel application scenario are significantly different from other traditional ways. In addition, the best effect of this novel application is related to the local climatic resources, and obtaining the relationship between the best heating effect and local climatic resources is helpful for the promotion of the novel solar heating building.

In this study, a novel solar building integrated with flat gravity heat pipes and phase change materials (BIHP-PCM) is proposed. Energy saving ratio and improvement degree of thermal discomfort are optimization goals of BIHP-PCM with HVAC and without HVAC, respectively. Thermophysical properties of PCM, including volumetric phase change enthalpy, phase change temperature, and thermal conductivity are optimized, and the best local heating effects are obtained at 61 cities in severe cold and cold regions of China. Taking the ratio of solar radiation to air temperature difference of the local heating season as the evaluation index of climatic resources, the relationship between the optimal heating performance of BIHP-PCM and the local climatic resources is studied, and the application potential zoning of BIHP-PCM is proposed. This study provides a new approach for the efficient utilization of PCM in solar heating buildings, and the conclusions on the PCM optimization (with/without HVAC) are helpful for forming a new understanding of how PCM plays a role in improving the indoor thermal environment. This study is of great significance for the application of BIHP-PCM to promote clean heating, reduce fossil energy consumption, and achieve the dual-carbon goal.

2. Model and methodology

2.1. Numerical model and validation

2.1.1. Numerical model of BIHP-PCM

The structure of the novel BIHP-PCM is shown in Fig. 1(a). Room A and Room B are two adjacent rooms with the same structure. The interior wall of BIHP-PCM which separates the two rooms is composed of the conventional building material and the PCM near the indoor environment, as shown in Fig. 1(b). The HP condensation section is closely embedded between the conventional building material and the PCM of the interior wall, and the HP evaporation section is covered with a heat-absorbing surface and is attached to the outer surface of the exterior wall. The HP has two different heat transfer mechanisms during the daytime and at nighttime, and the thermal resistance during the day can be only 170th of that at night [42]. During the day, the HP works normally and its equivalent thermal conductivity can reach 10^4 W/(m·K), transferring a mass of solar radiation heat absorbed at the evaporation section to the condensation section. Then the PCM efficiently stores heat transferred by the HP through heat conduction. Meanwhile, the interior wall also absorbs solar radiation heat that enters

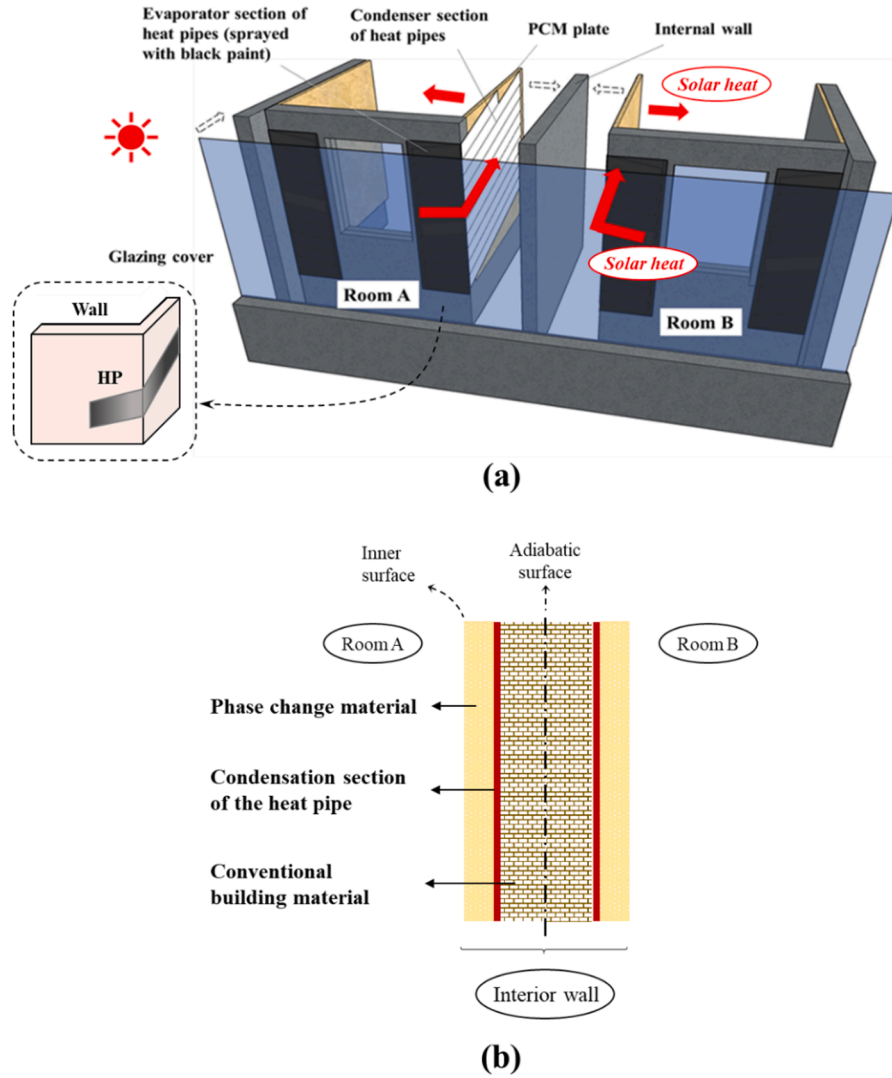


Fig. 1. Schematic diagram of BIHP-PCM: (a) overall structure; (b) section view of the phase change interior wall.

the room directly through the window. At night, the PCM releases heat to the indoor environment, and the HP is not in the highly efficient working mode, which avoids a large amount of heat dissipation from the PCM to the outdoor environment.

During the day, the temperature of the HP evaporation section is higher than that of the HP condensation section, and the heat transferred from the evaporation section to the condensation section can be obtained by:

$$Q_{HP, fw} = \frac{T_{eva} - T_{con}}{R_{fw}} = \frac{A_{sec} k_e (T_{eva} - T_{con})}{l_{eff}} \quad (1)$$

where $Q_{HP, fw}$ is the heat transferred by the HP in the forward direction (from the evaporation section to the condensation section), W; R_{fw} is the forward thermal resistance of the HP, K/W; T_{eva} and T_{con} are the temperatures of the evaporation section and the condensation section, respectively, °C; A_{sec} is the sectional area of the HP, m²; k_e is the equivalent thermal conductivity of the HP, which is valued as 2×10^4 W/(m·K) in forward heat transfer in the numerical model [36]; l_{eff} is the effective length of the HP, m, which is obtained by:

$$l_{eff} = \frac{l_{eva} + l_{con}}{2} + l_{adi} \quad (2)$$

where l_{eva} , l_{con} , and l_{adi} are the lengths of the evaporation section,

condensation section, and adiabatic section of the HP, respectively, m.

At night, the inner working fluid of HP is not able to form a flow circulation as the condensation section temperature is higher than the evaporation section temperature. The heat transfer model of HP can be regarded as heat conduction with three parallel thermal resistances of the two heat pipe walls and the working fluid [42]. The heat transferred from the condensation section to the evaporation section can be obtained by:

$$Q_{HP, rv} = \frac{T_{eva} - T_{con}}{R_{rv}} = (T_{eva} - T_{con}) \sum \frac{A_{sec, i} \lambda_i}{l_{eff}} \quad (3)$$

where $Q_{HP, rv}$ is the heat transferred by the HP in the reverse direction, W; R_{rv} is the reverse thermal resistance of the HP, K/W; $A_{sec, i}$ is the sectional area of the working fluid and the wall material, m²; λ_i is the thermal conductivity of the working fluid and the wall material, W/(m·K). The heat dissipation by the HP at night is about two orders of magnitude less than the heat gain during the day and is ignored in the numerical model.

The heat transfer model of the wall embedded with HP is shown in Fig. 2(a). The HP evaporation section absorbs heat under solar radiation, and most of the heat is transferred to the condensation section by the HP, while a small amount of heat is transferred to the exterior wall in contact with the evaporation section through thermal conduction. The HP condensation section embedded inside the interior wall continues to transfer heat to the interior wall, and this amount of heat can be divided

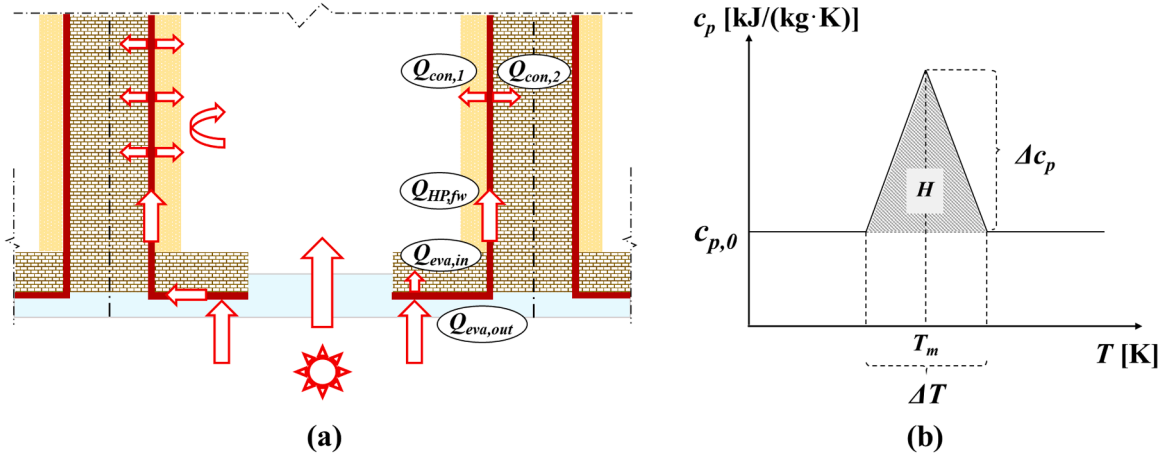


Fig. 2. Numerical model of BIHP-PCM: (a) heat transfer of the wall embedded with HP; (b) distribution of the equivalent specific heat of PCM.

into two parts on both sides. The heat balance of the above process can be expressed as follows.

$$Q_{eva,out} = Q_{HP,fw} + Q_{eva,in} = Q_{con,1} + Q_{con,2} + Q_{eva,in} \quad (4)$$

where $Q_{eva,out}$ is the effective heat transfer from the evaporation section of HP to the indoor environment, J; $Q_{eva,in}$ is the heat transfer from the evaporation section of HP to the exterior wall, J; $Q_{con,1}$ and $Q_{con,2}$ are the heat transfer from the condensation section of HP to the interior wall on two sides, J.

The specific heat capacity of PCM is treated by the equivalent specific heat method [43], as shown in Fig. 2(b). The equivalent specific heat capacity of PCM in the phase change process is treated as a triangular waveform curve distribution and is attached to the basic specific heat capacity $c_{p,0}$, which characterizes sensible heat. The difference of $c_{p,0}$ in solid and liquid phases is ignored in this model. The triangular waveform is symmetric about the phase change temperature T_m . The latent heat H is equal to the area of the shaded part, which is enclosed by the distribution curve and the basic specific heat capacity.

$$H = \frac{1}{2} \Delta T \Delta c_p \quad (5)$$

where H is the latent heat, kJ/kg; ΔT is the phase change temperature range, °C; Δc_p is the maximum change in specific heat capacity during the phase change process, kJ/(kg·K).

The available internal energy E_{int} of PCM is defined as:

$$E_{int} = \int_{T_{base}}^{T_{PCM}} m c_p(T) dT \quad (6)$$

Where, m is the total amount of PCM, kg; $c_p(T)$ is the equivalent specific heat capacity of PCM, J/(kg·K); T_{PCM} is the average temperature of PCM, °C; T_{base} is the base temperature for energy calculation, °C.

The melting fraction f of PCM is defined as:

$$f = \frac{\int_{T_1}^{T_{PCM}} c_p(T) dT}{\int_{T_1}^{T_2} c_p(T) dT} \quad (7)$$

where T_1 and T_2 are the lower and upper limits of the phase change temperature zone of PCM, respectively, °C.

The heat transfer model of the wall without HP is treated as conventional building walls, and the heat transfer equation and boundary conditions are as follows.

$$\rho_w c_{p,w} \frac{\partial T_w}{\partial \tau} = \lambda_w \frac{\partial^2 T_w}{\partial x^2} \quad (8)$$

$$-\lambda_w \frac{dT_w}{d\tau} \Big|_{x=0} = h_{out} (T_{out} - T_{w,out}) + q_{sol,out} \quad (9)$$

$$\lambda_w \frac{dT_w}{d\tau} \Big|_{x=d} = h_{in} (T_{in} - T_{w,in}) + q_{sol,in} \quad (10)$$

where ρ_w is the density of wall material, kg/m³; $c_{p,w}$ is the specific heat capacity of wall material, J/(kg·K); λ_w is the thermal conductivity of wall material, W/(m·K); h_{out} and h_{in} are the surface heat transfer coefficients at the outer and the inner surface of the wall, which are valued as 23.0 W/(m²·K) and 8.7 W/(m²·K), respectively; T_{out} and T_{in} are the outdoor air and the indoor air temperature, respectively, °C; $T_{w,out}$ and $T_{w,in}$ are the outer and the inner surface temperature of the wall, respectively, °C; $q_{sol,out}$ and $q_{sol,in}$ are the solar radiation absorbed at the outer and the inner surface of the wall, respectively, W/m².

The indoor air exchanges heat with the inner surface of building envelopes and is affected by air exchange and indoor heat disturbance. It is also affected by heating equipment in the case of rooms with HVAC. The energy balance equation of indoor air is as follows.

$$c_{p,a} \rho_a V_a \frac{dT_{in}}{d\tau} = \sum_{j=1}^6 h_{in,j} A_{w,j} (T_{w,in,j} - T_{in}) + K_{win} A_{win} (T_{gap} - T_{in}) + c_{p,a} \rho_a V_a ACH (T_{out} - T_{in}) + q_{int} + q_{HVAC} \quad (11)$$

where ρ_a is the air density, kg/m³; $c_{p,a}$ is the air specific heat capacity, J/(kg·K); V_a is the indoor air volume, m³; A_w and A_{win} are the surface area of the wall and the window, respectively, m²; K_{win} is the heat transfer coefficient of the window, W/(m²·K); T_{gap} is the air temperature in the glazing cover, °C; q_{int} is the indoor heat disturbance, W; q_{HVAC} is the heat input by heating equipment, W, which is zero for rooms without HVAC.

2.1.2. Model validation

An experimental study is used for model validation [16]. Two real-scale houses, a BIHP-PCM and a reference house (RB), are built in Beijing, as shown in Fig. 3. The length, width, and height of the two houses are all 2.4 m. The envelopes are all made of 150 mm rock wool board with no windows, and the dimension of the south glazing cover is 2.10 m × 1.50 m. A door with the same wall material is set on the east wall. The aluminum HP is filled with acetone as the working fluid and the filling ratio is 10 %. The areas of the evaporation section and condensation section of HP are 1.68 m² and 2.36 m², respectively. The surface of the HP evaporation section is covered with black paint. Calcium chloride hexahydrate of 0.09 m³ (153 kg) is applied as the PCM, which is encapsulated in boxes of 0.20 m × 0.15 m × 0.02 m. Both houses have no indoor heat disturbance. Air changes per hour is 0.1 h⁻¹. The experiment date is May 11 - 21.

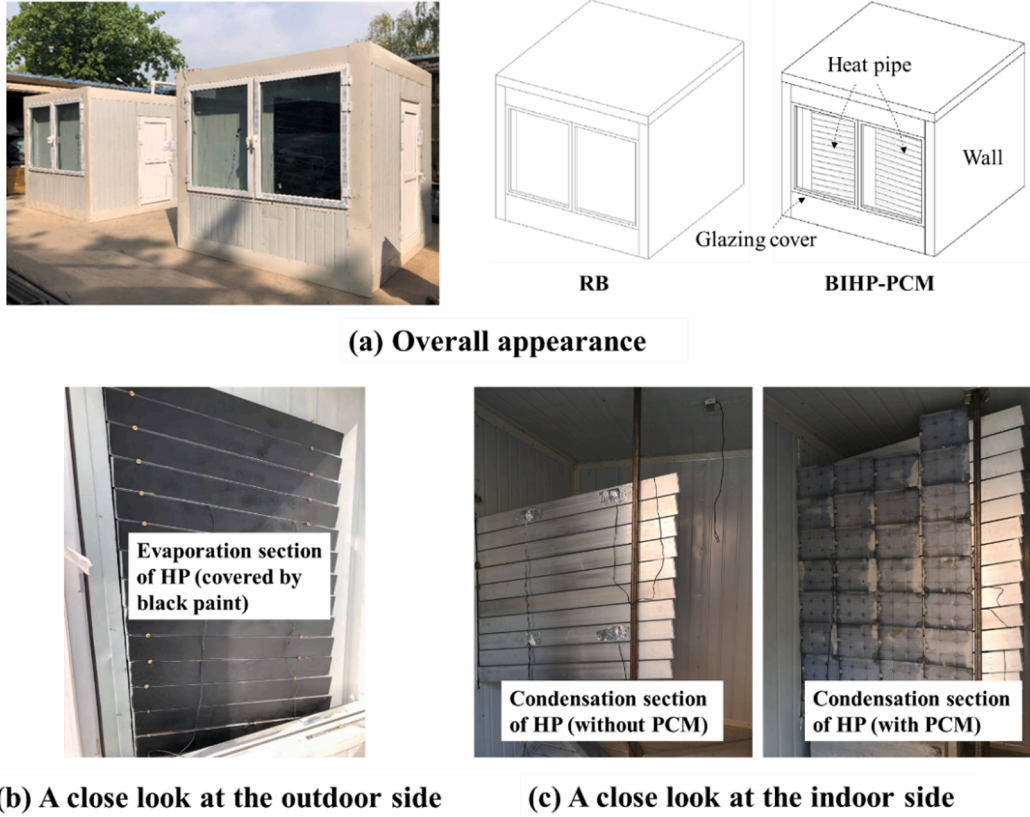


Fig. 3. Photograph of the experiment houses for model validation.

Fig. 4(a) shows the layout of temperature measuring points, including the indoor air temperature, temperatures of the HP evaporation and condensation sections, and temperatures of the inner and outer surfaces of the south wall. The indoor air temperatures at the height of 0.5 m, 1.0 m, and 1.5 m at the center of the two houses are recorded, and the natural room temperatures are represented by the mean value of the three temperatures. Climatic resources are recorded, including solar radiation intensity and outdoor air temperature. The temperatures are recorded by automatic recording thermometer (with an accuracy of $\pm 0.2^\circ\text{C}$, WZY-1, TJHY), and the solar radiation is recorded by automatic recording solar energy meter (with an accuracy of $\pm 10\text{ W/m}^2$, TES-132, TES). The recording time interval is 60 s.

The numerical model of BIHP-PCM is established in MATLAB and the finite difference method is adopted for calculation. Parameters such as room dimensions, thermophysical properties of wall materials and PCM, climatic resource conditions, heat pipe area, PCM amount, and ACH are input into the numerical model. The spatial step of the difference calculation is 5 mm and the time step is 60 s. May 21 is chosen as the simulation period, and May 11 - 20 is the pre-calculation period to eliminate the impact of the initial value. As shown in Fig. 4(b), the simulated room temperatures of the BIHP-PCM and the reference house on a day agree well with the experimental results, with R^2 higher than 0.98. The daily average temperature difference between the simulated and experimental results of the BIHP-PCM and the reference house is 0.3°C and 0.2°C , respectively. It should be noted that although the experiment is not conducted during the local heating season and the envelope settings are not the same as typical BIHP-PCM, only the input parameters including climatic conditions, heat pipe area, and PCM amount need to be adjusted in the numerical model to simulate a typical BIHP-PCM, and the simulation reliability of the heat transfer process is not affected. Therefore, subsequent calculations in the present study are based on this numerical model.

2.2. Optimization index and method

2.2.1. Optimization index

As the terminal of heat storage and release in a BIHP-PCM, the interior envelope combined with PCM has an important effect on the indoor thermal environment. The indoor thermal environment of the BIHP-PCM with or without HVAC is completely different, and to optimize the thermophysical properties of the PCM, reasonable heating effect evaluation indexes should be adopted as optimization objectives respectively. The optimal heating effects of the optimized BIHP-PCM at different cities with different climatic resources are various, so it is necessary to select an appropriate local climatic resource evaluation index, so as to pre-estimate the heating effects of the optimized BIHP-PCM at unstudied cities based on local climatic resources.

(a) Heating effect without HVAC

For the BIHP-PCM without HVAC, the indoor environment is not regulated by heating equipment, and the heating energy consumption is zero. However, the room temperature is uncontrolled and may be lower than the lower limit of the thermal comfort temperature zone, resulting in indoor thermal discomfort. The integrated discomfort degree of cold of the heating season *IDDC* proposed in a previous study is used as the evaluation index of the indoor thermal environment of the BIHP-PCM [41], and the definition is as follows:

$$IDDC = \int_{\text{heating season}} (T_L - T_{op}) d\tau, \quad T_{op} < T_L \quad (12)$$

where *IDDC* is the integrated discomfort degree of cold, K-h; τ is time, h; T_L is the lower temperature limit value of the thermal comfort zone, $^\circ\text{C}$; T_{op} is the indoor operative temperature, $^\circ\text{C}$, which is obtained as:

$$T_{op} = \frac{h_c T_{in} + h_r T_r}{h_c + h_r} \quad (13)$$

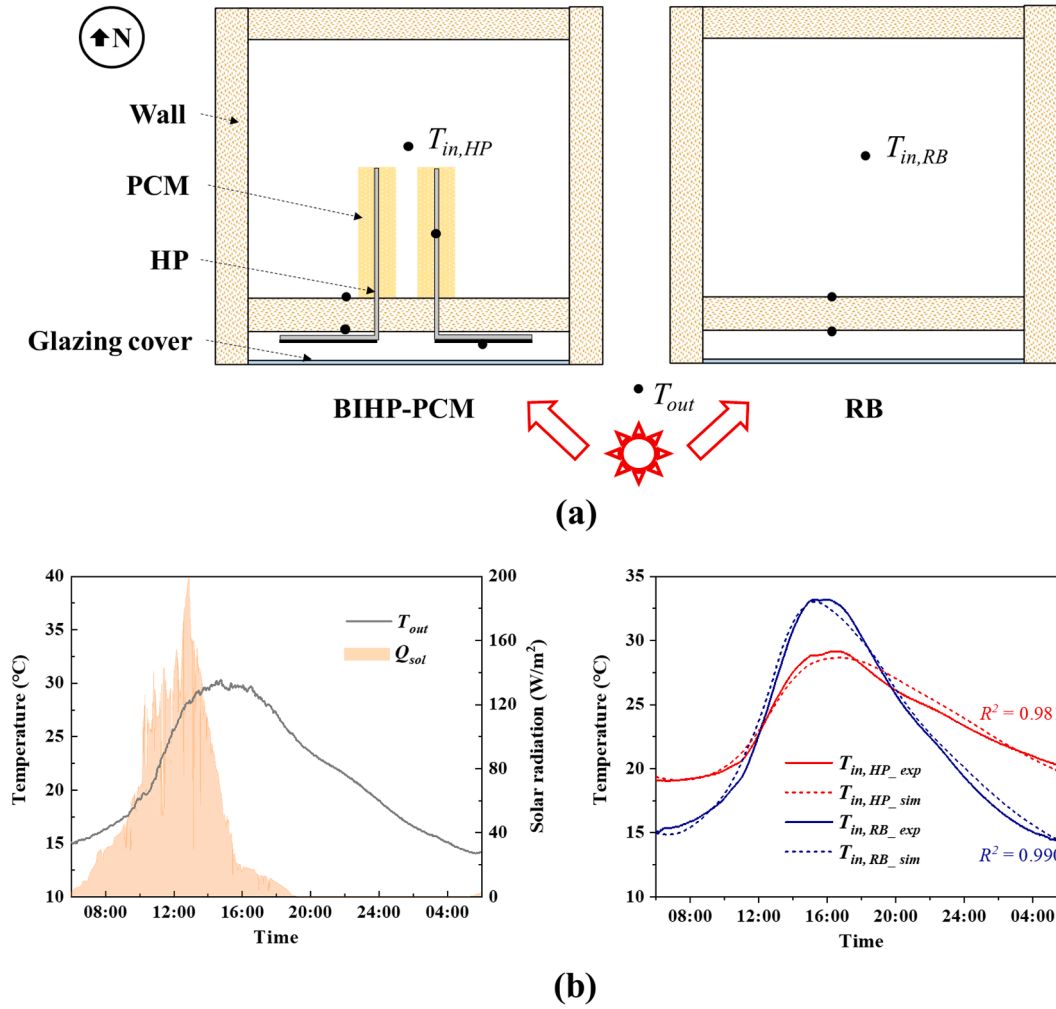


Fig. 4. Model validation: (a) top view of the layout of the temperature measuring points; (b) climatic resources, simulated and experimental room temperatures.

where T_{in} is the indoor air temperature, °C; h_r and h_c are the radiative and convective heat transfer coefficients, respectively, W/(m²·K); T_r is the mean radiant temperature, °C, which can be calculated by:

$$T_r = \frac{\sum_{j=1}^6 A_j T_{wall,inn,j} + A_{win} T_{win}}{\sum_{j=1}^6 A_j + A_{win}} \quad (14)$$

where $T_{wall,inn,j}$ is the inner surface temperature of each wall, °C; T_{win} is the inner surface temperature of the window, °C; A_j is the inner surface area of each wall, m²; A_{win} is the window area, m².

The *IDDC* at different cities is difficult to compare directly as the climatic resource conditions and the periods of local heating seasons are very different. The improvement degree of thermal discomfort *IDTD* is used to describe the reduction of thermal discomfort of BIHP-PCM compared to the reference building at the same location, which is the same building as the BIHP-PCM except not equipped with HPs and PCMs.

$$IDTD = \frac{IDDC_{RB} - IDDC_{HP}}{IDDC_{RB}} \times 100\% \quad (15)$$

where $IDDC_{RB}$ and $IDDC_{HP}$ are the integrated discomfort degree of cold of the reference building and the BIHP-PCM, respectively, K·h.

It can be seen that the *IDTD* of the BIHP-PCM without HVAC is between 0 % and 100 %, and the smaller the $IDDC_{HP}$, the larger the *IDTD*, which is the optimization goal.

(b) Heating effect with HVAC

For the BIHP-PCM with HVAC, the heating equipment operates to maintain the room temperature not below the lower temperature limit of the thermal comfort zone. The heating energy consumption of HVAC is different for the BIHP-PCM with different thermophysical properties. The cumulative heating energy consumption of the local heating season Q is an index to evaluate the thermal performance of the BIHP-PCM, and is as follows:

$$Q = \int_{\text{heating season}} q(\tau) d\tau \quad (16)$$

where Q is the cumulative heating energy consumption of the local heating season, J; $q(\tau)$ is the heating load of the room, W.

The energy saving ratio *ESR* is used to describe the energy saving degree of the BIHP-PCM compared with the reference building without HPs and PCMs at the same location.

$$ESR = \frac{Q_{RB} - Q_{HP}}{Q_{RB}} \times 100\% \quad (17)$$

where Q_{RB} and Q_{HP} are the cumulative heating energy consumption of the reference building and the BIHP-PCM, respectively, J.

It can be seen that the *ESR* of the BIHP-PCM with HVAC is between 0 % and 100 %, and the smaller the Q_{HP} , the larger the *ESR*, which is the optimization goal.

(c) Local climatic resources

The optimal heating performance of BIHP-PCM is closely related to

local climatic resources. The ratio of solar radiation to temperature difference describes the maximum potential benefit of solar radiation resources in creating an indoor environment with a certain temperature difference at the average local outdoor temperature [44]. Considering the different heating demand periods at different cities, the local heating season is taken as the calculation period, and the ratio of solar radiation to temperature difference of the heating season $RRTD_{HS}$ is defined as follows for a south-facing building:

$$RRTD_{HS} = \frac{Q_{sol,ave}}{T_{set} - T_{out,ave}} \quad (18)$$

where $Q_{sol,ave}$ is the average southern solar radiation throughout the heating season, W/m^2 ; T_{set} is the setting temperature of indoor heating, $^{\circ}C$; $T_{out,ave}$ is the average outdoor air temperature throughout the heating season, $^{\circ}C$.

2.2.2. Optimization method

In this study, particle swarm optimization is used as a multi-parameter optimization method based on the inverse problem idea. The PCM inside the interior wall of the BIHP-PCM is arranged in a fixed position, which is between the condensation section of the HP and the indoor air. To reduce optimization variables and further focus on thermophysical properties, the phase change temperature range and thickness of PCM remain unchanged in this study, and the volumetric enthalpy ρH , phase change temperature T_m , and thermal conductivity λ are three independent parameters to be optimized. The three thermophysical properties, whose values are limited by the set ranges, constitute an enclosed three-dimensional space, where a coordinate represents a combination of thermophysical properties of the PCM. Considering the time and calculation cost, the population of the particle swarm is set to be 20 and the total number of iterations is 30.

The solving process of the particle swarm optimization algorithm can be concluded as the following six steps [36].

- (1) Initialize: The initial positions of individuals are randomly assigned in the closed three-dimensional space.

- (2) Calculate: The objective parameters ($IDDC$ without HVAC, Q with HVAC) of all particles are simulated by the BIHP-PCM numerical model with the spatial coordinate (ρH , T_m , λ).
- (3) Select: The particle that makes the objective parameter optimal in the whole particle swarm is found in each iteration.
- (4) Update: In the process of optimization, each particle has a search velocity in every variable dimension, which guides the particle to move to the position of the optimal particle in the solution space, and the positions and velocities of all particles are adjusted as the iteration progresses.
- (5) Iterate: Repeat step (2) - step (4) as all particles gradually cluster around the optimal position.
- (6) Output: The solving process ends as the iteration number reaches the setting value, and the ρH , T_m , λ corresponding to the optimal position are output as the optimal thermophysical properties of PCM.

2.3. Simulation settings

2.3.1. City selection

China is categorized into five climate zones according to the building thermal design zoning [45]: severe cold, cold, hot summer and cold winter, hot summer and warm winter, and moderate. The severe cold zone is further categorized into severe cold A, severe cold B and severe cold C, and the cold zone is further categorized into cold A and cold B. 61 cities from the severe cold and cold zones are selected in this study, as shown in Fig. 5. The selection principles are: (1) evenly geographically distributed; (2) provincial capital cities and big cities with large populations are preferred; (3) no more than five cities in a province. The climate data of these cities is from DeST, a building simulation software.

2.3.2. Building information

A south-facing room of a multi-story BIHP-PCM is considered for simulation. All the envelopes are interior envelopes except the south wall, and the heat transfer between the simulated room and adjacent rooms is negligible. The simulated room is 4 m long, 3 m wide, and 3 m high. The window-wall ratio of the south wall is 0.3. The ACH is set to be 0.5 and 5.0 when the operative temperature T_{op} is under and above 26



Fig. 5. 61 selected cities for simulation.

°C, respectively. The HP evaporation section is attached to the outer surface of the south wall, and the east and west walls are embedded with the HP condensation section. The conventional building material of the east and west interior walls is 12 cm thick brick, and the PCM layer is 6 cm thick. The solar radiation transmittance of the window is 0.75. The absorptance of the heat-absorbing surface of the BIHP-PCM and that of the outer surface of the south wall of the reference building is 0.9. As shown in Table 1, the exterior envelope with a high thermal resistance meets the relevant requirement of the corresponding building climate zone according to a Chinese national standard [46]. The lower temperature limit of the thermal comfort zone T_L is set as 18 °C, which is the minimum room temperature guaranteed by the heating equipment of BIHP-PCM with HVAC.

Table 2 lists the thermophysical properties of some PCMs used in buildings and studied in the literature. In this study, the value ranges of the three parameters ρH , T_m , and λ are 0–420 MJ/m³, 10–30 °C, and 0–10 W/(m·K), respectively. The phase change temperature range ΔT is given 2 °C in the present study to ignore the effect of phase change temperature range and focus on analyzing the influence of T_m . The PCM of the non-optimized BIHP-PCM is potassium fluoride tetrahydrate, and the reference building is not equipped with HPs and PCMs but is composed of bricks of the same geometric size.

3. Results and discussion

The BIHP-PCM greatly improves the indoor thermal environment compared to the reference building, but it may not guarantee thermal comfort throughout the entire heating season, in which case HVAC is required. For the BIHP-PCM with or without HVAC, the indoor thermal environment evaluation indexes and optimization objectives are significantly different, and the PCM optimization is carried out respectively. The optimal thermophysical properties, i.e., ρH , T_m , and λ of PCM of the two kinds of BIHP-PCM are investigated and compared. Based on the optimized PCM design, the optimal heating performance of BIHP-PCM with and without HVAC is analyzed, and the relationship between the optimal heating effect and the evaluation index of local climatic resources is obtained.

3.1. Optimization of PCM: without HVAC

The natural room temperature is not controlled and there may be thermal discomfort for a BIHP-PCM without HVAC, so the optimization aims to minimize *IDDC*. The optimal PCM design can be given by the proposed particle swarm optimization. Fig. 6 shows the optimal ρH , T_m , and λ of PCM and corresponding *IDDC* during all iterations in the optimization process, taking Tianjin as an example. In the first seven iterations, the optimal ρH significantly increased, and the optimal *IDDC* sharply decreased. Subsequently, the optimal λ rises, and the optimal T_m fluctuates around 19.0 °C and remains unchanged for the last ten iterations. Therefore, the optimal thermophysical properties of PCM are ($\rho H = 420$ MJ/m³, $T_m = 19.0$ °C, $\lambda = 6.3$ W/(m·K)) in Tianjin, and the

Table 1
Heat transfer coefficients of the exterior building envelope in this study.

Building climate zone		Wall construction	K_{wall} (W/(m ² ·K))	K_{win} (W/(m ² ·K))
Severe cold	Severe cold A	Polystyrene board (100 mm) + Concrete hollow block (240 mm)	0.25	1.6
	Severe cold B	Polystyrene board (80 mm) + Concrete hollow block (240 mm)	0.25	1.8
	Severe cold C	Polystyrene board (60 mm) + Concrete hollow block (240 mm)	0.30	2.0
	Cold	Polystyrene board (60 mm) + Concrete hollow block (240 mm)	0.36	2.2
Cold	Cold A	Polystyrene board (60 mm) + Concrete hollow block (240 mm)	0.36	2.2
	Cold B	Polystyrene board (60 mm) + Concrete hollow block (240 mm)	0.36	2.2

Table 2
Thermophysical properties of some PCMs [47].

PCM	λ (W/(m·K))	ρH (MJ/m ³)	T_m (°C)
CaCl ₂ ·6H ₂ O	1.09(solid)/0.54(liquid)	289	27
n-Octadecane	0.36(solid)/0.15(liquid)	188	27
PEG600	0.19	141	22
RT27	0.2	145	27
SP25A8	0.6	248	25
KF·4H ₂ O [48]	0.61(solid)/0.48(liquid)	333	19
L30 [49]	1.02(solid)/0.56(liquid)	420	30
S27 [49]	0.79(solid)/0.48(liquid)	323	27
TM/EG [50]	9.72	134	23

minimum *IDDC* is 192 K·h, which is the best utilization achieved by BIHP-PCM.

The heating effect of the optimized BIHP-PCM is calculated based on the optimal PCM. Fig. 7 shows the indoor operative temperature of optimized BIHP-PCM ($T_{op,PCM,OPT}$), initial non-optimized BIHP-PCM ($T_{op,PCM,INI}$), and the reference building ($T_{op,RB}$) throughout the whole heating season, and the outdoor air temperature (T_{out}) and solar radiation intensity (Q_{sol}) in Tianjin. Non-optimized BIHP-PCM transfers a mass of solar heat to the indoor environment, and the average $T_{op,PCM,INI}$ is 18.4 °C. However, the resistance to extreme weather is still poor, leading to the lowest $T_{op,PCM,INI}$ of only 8.5 °C in early December and mid-January. *IDDC*_{PCM,INI} decreases to 2206 K·h, and *IDTD*_{PCM,INI} is 87.4 %. The average $T_{op,PCM,OPT}$ of the optimized BIHP-PCM is 20.1 °C and the minimum value is 13.9 °C, and $T_{op,PCM,OPT}$ is below the thermal comfort zone only in a few days in mid-January. *IDDC*_{PCM,OPT} decreases to 192 K·h, and *IDTD*_{PCM,OPT} is 98.9 %. The results show that the indoor thermal environment of the BIHP-PCM without HVAC is significantly improved by the effective use of solar energy, and the heating effect can be further improved by optimizing the thermophysical properties of PCM.

The effects of ρH and T_m on *IDDC* and *IDTD* of the whole heating season are shown in Fig. 8, with λ valued as 0.5 W/(m·K). With ρH and T_m of 50 MJ/m³ and 22 °C respectively, *IDDC* is 5008 K·h and *IDTD* is 71.5 %. With ρH and T_m of 420 MJ/m³ and 19 °C respectively, *IDDC* is 1924 K·h and *IDTD* is 89.0 %. *IDDC* decreases with the increase of ρH and decreases with T_m approaching 19 °C. The results show that increasing ρH of PCM is beneficial to solar heat storage, and T_m should be about 0–2 °C above the lower limit of the thermal comfort temperature zone.

The simulation shows that the BIHP-PCM can realize zero carbon heating at 22 out of 61 selected cities, and there is more than one PCM design to reach zero carbon heating, which is not further discussed in this study. For the other 39 non-zero carbon heating cities, there is only one optimal PCM design that minimizes *IDDC* at each location. The optimal results of all the non-zero carbon heating cities indicate that the optimal ρH are all the maximum value, which is 420 MJ/m³, and the optimal λ range from 5 to 9 W/(m·K), as shown in Fig. 9. It indicates that the interior wall should be of high thermal conductivity and the largest heat capacity. The optimal T_m are various and most are within 1 °C above the lower temperature limit of the thermal comfort zone T_L . The optimal T_m at some locations is lower than T_L , which is due to the bottom-supporting effect of PCM on room temperature. These locations are of poor solar radiation resources and *RRTD*_{HS} lower than 5.7 W/(m²·K).

Taking Karamay as an example, the optimal T_m is 13.8 °C and the corresponding *IDDC* is 9424 K·h, while *IDDC* is 11,650 K·h with T_m taken as 19 °C. As can be seen from the temperature curves in Fig. 10, although the room temperature can be maintained in the thermal comfort zone before mid-December and after mid-February when T_m is 19 °C, T_{op} is significantly lower than T_m between mid-December and early February with low solar radiation. The PCM has been completely solidified and is unable to exert the advantages of heat storage and release in the phase change process. The lowest T_{op} even reaches 0 °C. On the contrary, although PCM with T_m of 13.8 °C does not undergo the phase change process until the room temperature drops below the thermal comfort

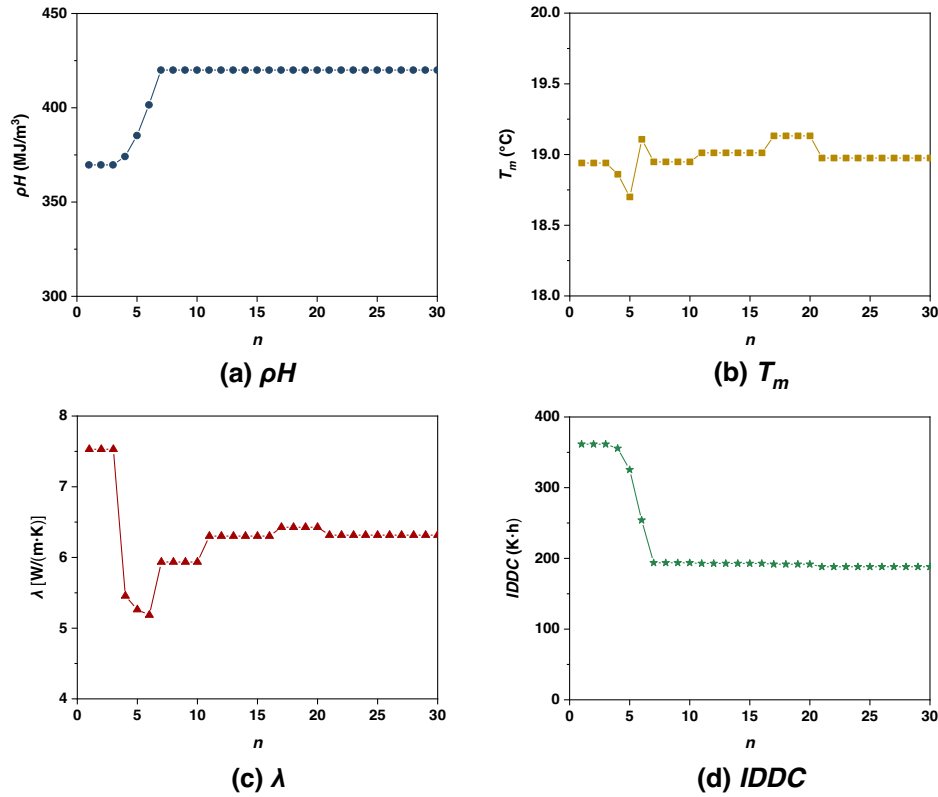


Fig. 6. Optimal parameters in the optimization process (Tianjin, without HVAC).

zone, it helps to maintain the room temperature near the phase change temperature for a long time when solar radiation is poor. In this way, a bottom temperature platform is built to prevent the deterioration of T_{op} , as shown in the circle, forming the bottom-supporting effect of PCM on room temperature. It should be noted that the $RRTD_{HS}$ at this city is only $5.52 \text{ W}/(\text{m}^2 \cdot \text{K})$ and the local solar energy resources are generally weak, thus there is a lack of sufficient solar radiation heat to support the phase change of PCM at a higher temperature near 18°C to reduce $IDDC$. However, for cities with higher $RRTD_{HS}$ and more abundant solar radiation resources, a higher T_m can maintain room temperature at a higher level for a long period, which is more beneficial to indoor thermal comfort [51].

3.2. Optimization of PCM: with HVAC

Although BIHP-PCM reduces indoor thermal discomfort significantly, it still may not meet the indoor thermal requirements during all the heating season, in which case HVAC is needed. The indoor thermal comfort can be guaranteed with HVAC, but it will lead to heating energy consumption, so the optimization goal is to minimize the cumulative heating energy consumption of the heating season Q .

Taking Tianjin as an example, the optimized thermophysical properties of PCM are ($\rho H = 420 \text{ MJ}/\text{m}^3$, $T_m = 19.5^\circ\text{C}$, $\lambda = 6.8 \text{ W}/(\text{m} \cdot \text{K})$), and the minimum Q is 68 MJ , which is the best performance achieved by BIHP-PCM. Fig. 11 shows the indoor operative temperature of optimized BIHP-PCM, initial non-optimized BIHP-PCM, and the reference building during the whole heating season in Tianjin. In periods of weak solar radiation, such as late November and mid-January, the reference building and the non-optimized BIHP-PCM both need HVAC in operation for a long time to control the room temperature not lower than 18°C . However, due to the large amount of heat released by PCM near the phase change temperature, the natural operative temperature of the optimized BIHP-PCM is obviously increased, and heating by HVAC is needed only in a few days around Jan. 14. During the period from late

January to late February, there are many cases in which the room temperature of the reference building reaches 18°C , while that of the BIHP-PCM still maintains above the lower limit of the thermal comfort temperature zone. In particular, when the room temperature of the optimized BIHP-PCM approaches 18°C several times, a large amount of solar radiation heat is introduced indoors the next day and the $T_{op,PCM,OPT}$ rises again, so the HVAC hardly needs to run. The average $T_{op,PCM,INI}$ and the average $T_{op,PCM,OPT}$ are 0.9°C and 1.5°C higher than the average $T_{op,RB}$, respectively, indicating that the average room temperature level of the BIHP-PCM in the whole heating season is significantly improved because of the effective utilization of solar energy.

Fig. 12(a) shows the heating load curves of the three houses during the heating season. The green curve and the orange curve have the longest period and the shortest period with the value greater than 0, respectively, indicating the HVAC operating time of the reference building and the optimized BIHP-PCM are the longest and the shortest, respectively. Fig. 12(b) shows the total heating energy consumption Q , operating time of HVAC τ_{on} , and average heating load of the operating time per unit area q'_a . Q is the product of τ_{on} , q'_a , and the room area. Q of the reference building is 1220 MJ , and Q of the non-optimized BIHP-PCM and the optimized BIHP-PCM are 341 MJ and 68 MJ , respectively, and ESR are 72.1% and 94.4% , respectively. The τ_{on} of the reference building, the non-optimized BIHP-PCM, and the optimized BIHP-PCM are 1475 h , 735 h , and 131 h , respectively, and the q'_a are $19.2 \text{ W}/\text{m}^2$, $10.7 \text{ W}/\text{m}^2$, and $12.1 \text{ W}/\text{m}^2$, respectively. It indicates that the significant reduction of Q is due to the large decrease of both τ_{on} and q'_a . As for τ_{on} , the high thermal conductivity and thermal diode of HP help to introduce a large amount of solar heat unidirectionally into the room, and then the PCM efficiently stores heat through heat conduction. Hence, the natural room temperature of BIHP-PCM maintains above 18°C much longer than the reference building, and the HVAC needs to work for less time. As for q'_a , since the PCM has a phase change temperature range (2°C), when the room temperature reaches 18°C and the HVAC is in operation, the PCM temperature with T_m of 19.5°C is not

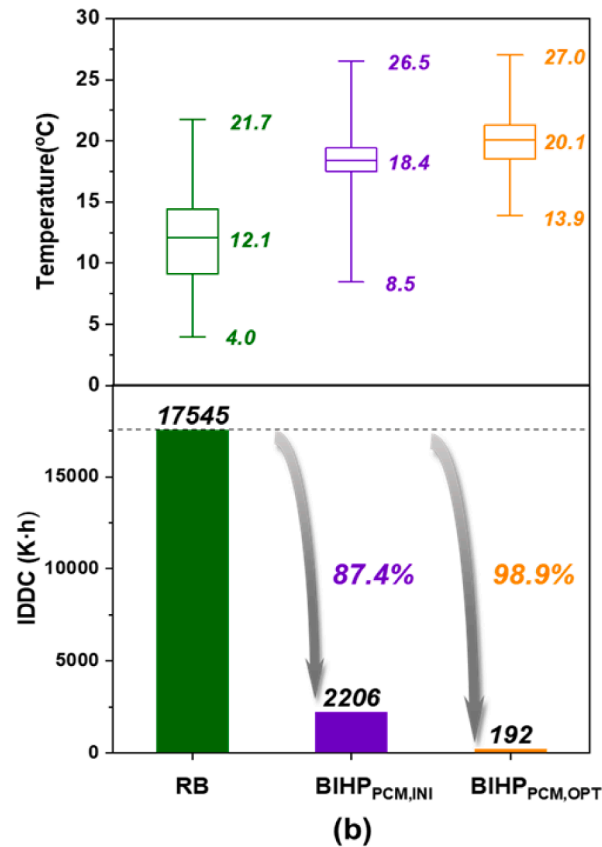
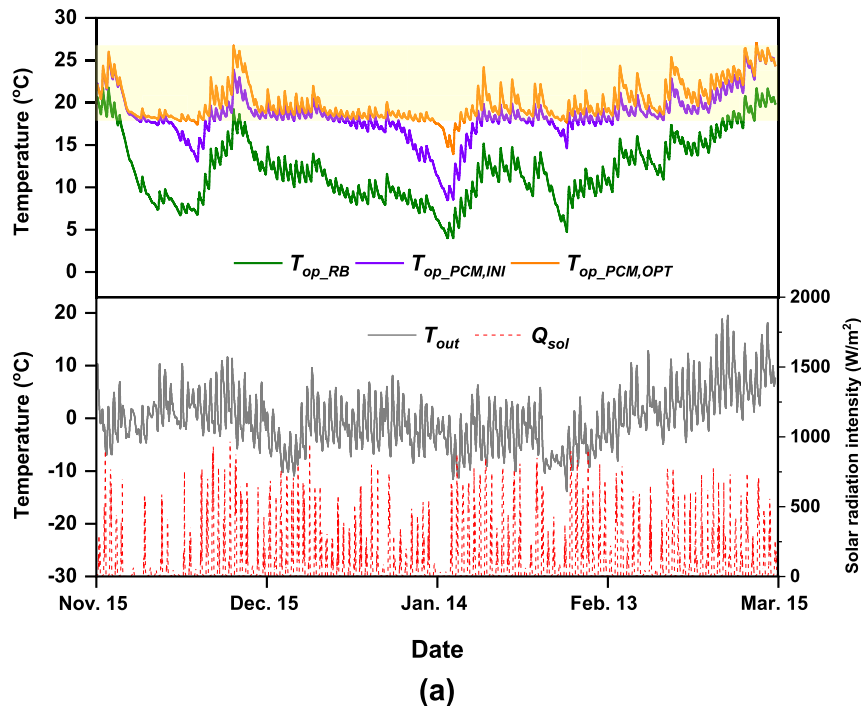
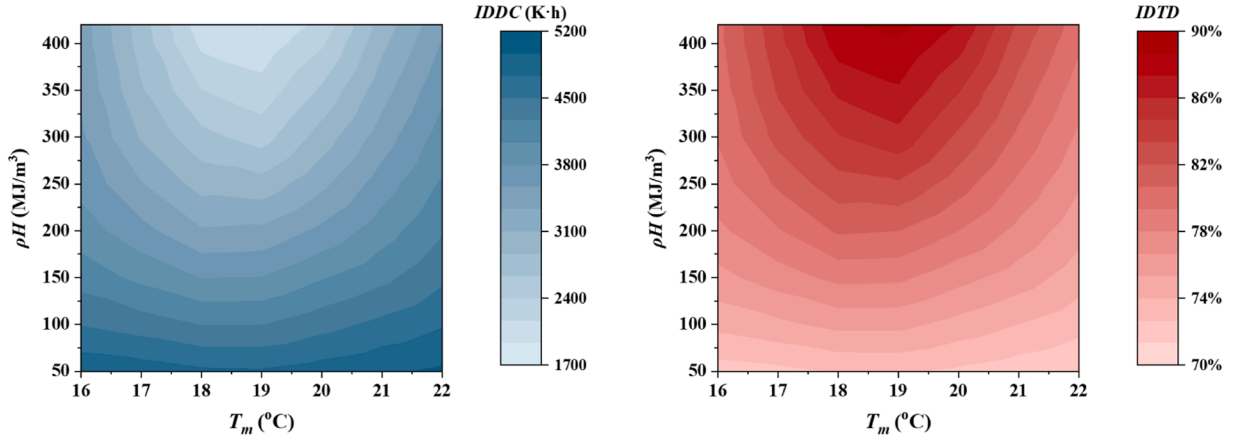
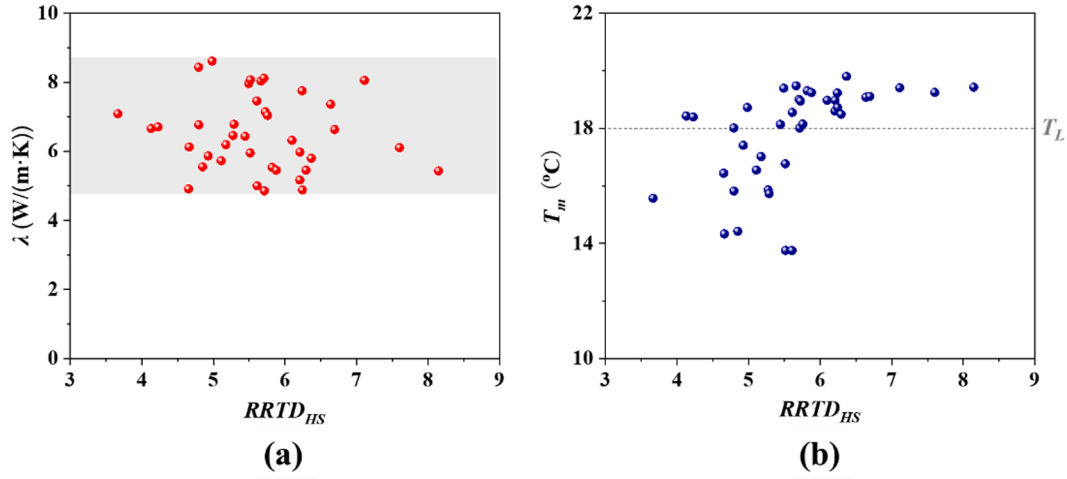
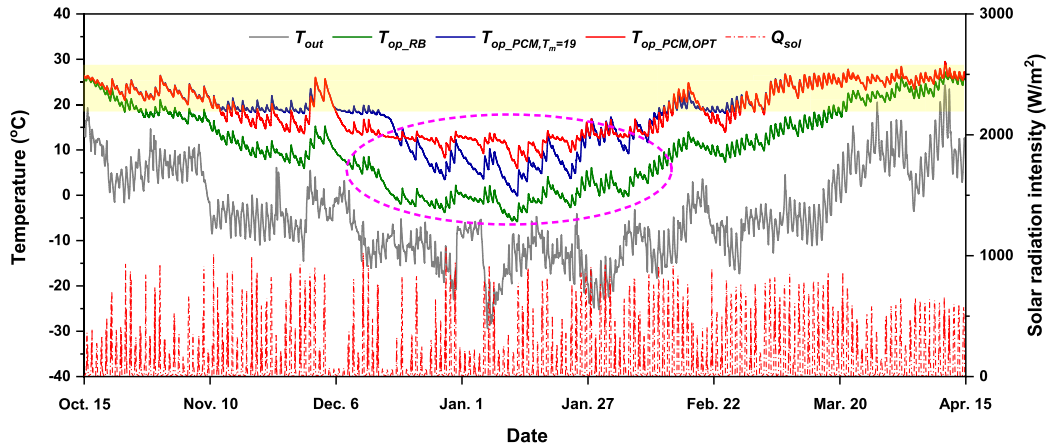


Fig. 7. (a) T_{op} curve graph of different houses; (b) T_{op} box plot and IDDC reduction (Tianjin, without HVAC).

lower than the lower limit of the phase change temperature zone, which is 18.5 $^{\circ}\text{C}$. Therefore, the PCM is still releasing latent heat into the room, which leads to a significant reduction in q'_a .

To further investigate the optimized phase change interior wall in improving the indoor thermal environment, Nov. 17 - 22 is studied as an

example. Solar radiation is rich during Nov. 17 - 19 and weak during Nov. 20 - 22. Climatic resources of the studied period are shown in Fig. 13(a), including outdoor air temperature T_{out} and solar radiation Q_{sol} . The wall surface temperature, releasing heat flux, available internal energy, and melting fraction of PCM with different ρH , T_m , and λ are

Fig. 8. Influence of ρH and T_m on IDDC and IDTD (Tianjin).Fig. 9. Optimal thermophysical properties at non-zero carbon heating cities (without HVAC): (a) λ and (b) T_m .Fig. 10. Comparison of T_{op} in different houses (Karamay, without HVAC).

simulated. The two variables that are not the research variable in this case are set to be the optimal values, that is, ρH , T_m , and λ are respectively taken as 420 MJ/m³, 19.5 °C, and 6.8 W/(m·K). The base temperature of available internal energy is slightly lower than the lowest PCM temperature and is set as 12 °C.

Fig. 13(b) shows the thermal performance with different phase change temperature T_m . The surface temperature, heat flow, available

internal energy, and melting fraction of the wall with T_m of 17.0 °C (green line), 19.5 °C (blue line), and 22.0 °C (red line) during the studied period are shown respectively. The PCM with T_m of 22.0 °C starts to release latent heat on Nov. 19 and the surface temperature is stable at around 22.0 °C, however, this heat release moment is too early since the room temperature is still several degrees higher than 18 °C. The phase change temperature zone of the PCM with T_m of 17.0 °C is all lower than

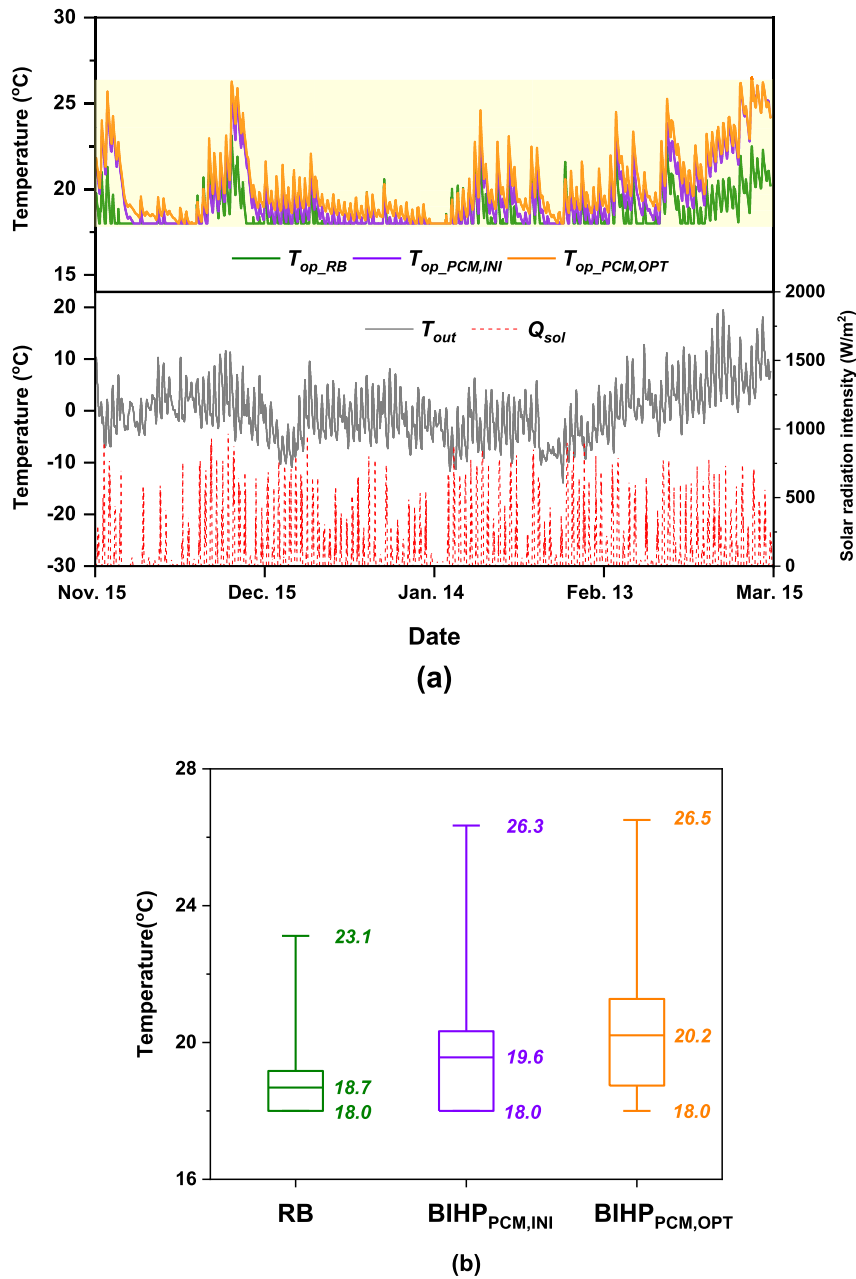


Fig. 11. Comparison of T_{op} in different houses (Tianjin, with HVAC): (a) curve graph and (b) box plot.

the thermal comfort zone and melting fraction f remains as 1 all the time, except the wall temperature drops to the phase change temperature zone on Nov. 22 due to radiative heat transfer with the inner surface of the exterior wall, and only about 1.1 % of the latent heat is released. The moment of the phase change process with T_m of 19.5 °C is between the other two, and a large amount of latent heat is released to maintain indoor thermal comfort when the room temperature is close to 18 °C, which is the most suitable choice of T_m .

Fig. 13(c) shows the thermal performance of the interior wall with different volumetric enthalpy ρH . In the first three days when solar radiation is abundant, the PCM temperature is always higher than its phase change temperature zone and remains in the liquid state, so the surface temperature fluctuation is obvious as the PCM uses sensible heat for heat storage and release. For the PCM with the smallest ρH (green line), the surface temperature and the heat flux decrease significantly on Nov. 22, as the PCM has already released all the latent heat and melting fraction f drops to 0. On the contrary, the f with ρH of 200 MJ/m³ (blue

line) and 420 MJ/m³ (red line) at the last moment of the studied period are 0.27 and 0.62, respectively, meaning the latent heat still has not been completely released.

Fig. 13(d) shows the thermal performance with different thermal conductivity λ . The wall with λ of 0.5 W/(m·K) shows the lowest surface temperature and heat flux. In the first three days, the PCM remains in the liquid state and the surface temperature fluctuation is obvious. In the last three days, the PCM temperature decreases to the phase change temperature zone as solar radiation is very weak, so latent heat is released indoors with the phase change process of PCM, and E_{int} decreases rapidly and f decreases from 1. However, the thermal performance with different λ is close, especially for PCM with a large λ .

From the above analysis, it can be seen that the reasonable selection of ρH and T_m of PCM is crucial to the heating performance of BIHP-PCM. Taking λ as 0.5 W/(m·K), the effects of ρH and T_m on Q and ESR of the whole heating season are shown in Fig. 14. With ρH and T_m of 50 MJ/m³ and 22 °C respectively, Q is 581 MJ and ESR is 52.4 %. With ρH and T_m of

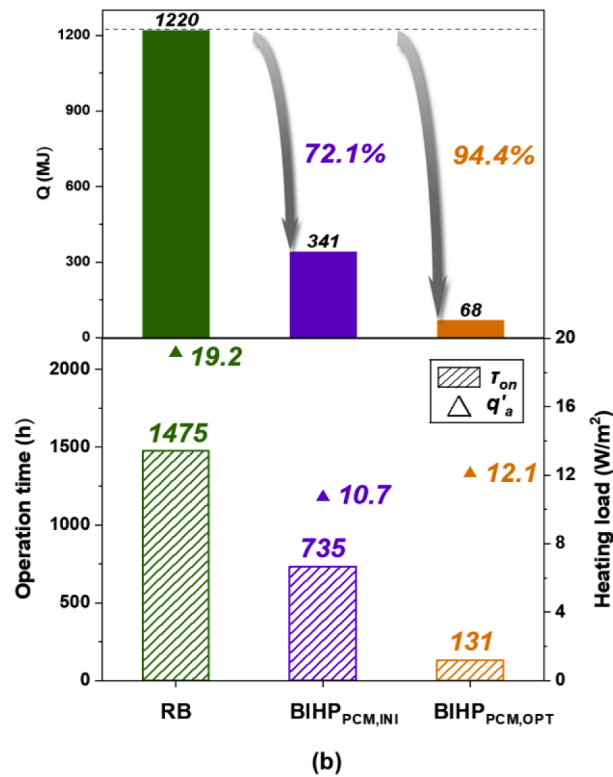
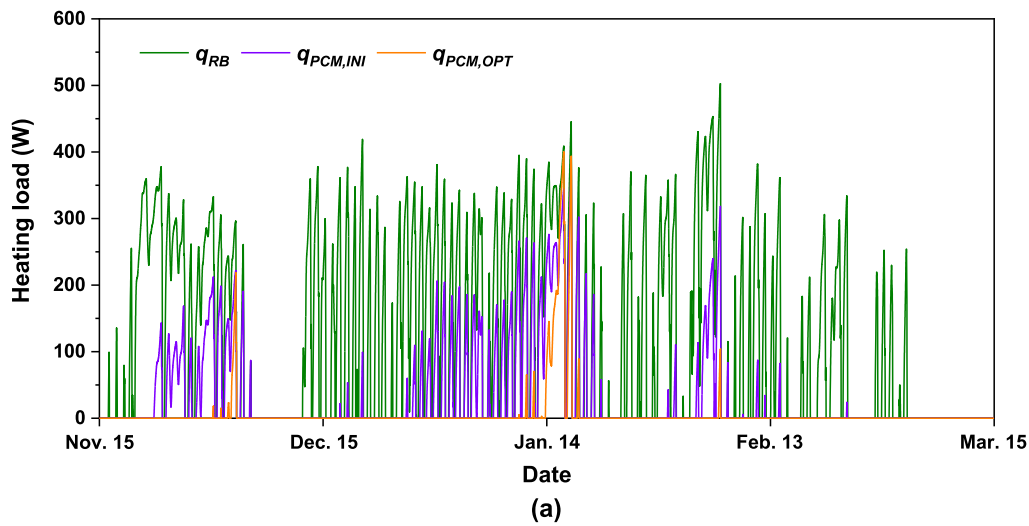


Fig. 12. (a) Curve graph of heating load; (b) heating energy consumption, operating time of HVAC, and average heating load (Tianjin, with HVAC).

420 MJ/m³ and 20 °C respectively, Q is 266 MJ and ESR is 78.2 %. Q decreases with the increase of ρH and decreases with T_m approaching 20 °C. On the one hand, if T_m is too high, the PCM releases latent heat too early when the room temperature is still significantly higher than the lower limit of the thermal comfort temperature zone T_L , resulting in heating energy consumption as the room temperature continues to decrease later. On the other hand, if T_m is too low so that the phase change temperature zone is completely below T_L , PCM remains in the liquid state during the whole heating season and its advantage of massive heat capacity during the phase change process is completely unexploited. In summary, the increase of ρH is conducive to the storage of solar radiation heat, and T_m should be above T_L with a certain temperature difference (1–2 °C) to release the stored heat in time as the room temperature is about to drop to 18 °C. This is consistent with the

results of BIHP-PCM without HVAC, as presented in Section 3.1.

By comparing Fig. 8 and Fig. 14, it can be seen that when the phase change temperature zone of PCM is completely below T_L , changing T_m has no effect on the heating effect of the BIHP-PCM with HVAC, but the heating effects of different T_m are different without HVAC. It is because the uncontrolled room temperature without HVAC may be lower than 18 °C, and T_m has an impact on the indoor environment. However, the room temperature keeps above 18 °C with HVAC, therefore, the latent heat distributed below the thermal comfort zone is not used.

The optimal PCM thermophysical properties of the non-zero carbon heating cities show that the optimal ρH are all the maximum value, which is 420 MJ/m³, and the optimal λ range from 3.5 to 9.0 W/(m·K), as shown in Fig. 15. The results indicate that the optimal interior wall should be of high thermal conductivity and the largest heat capacity,

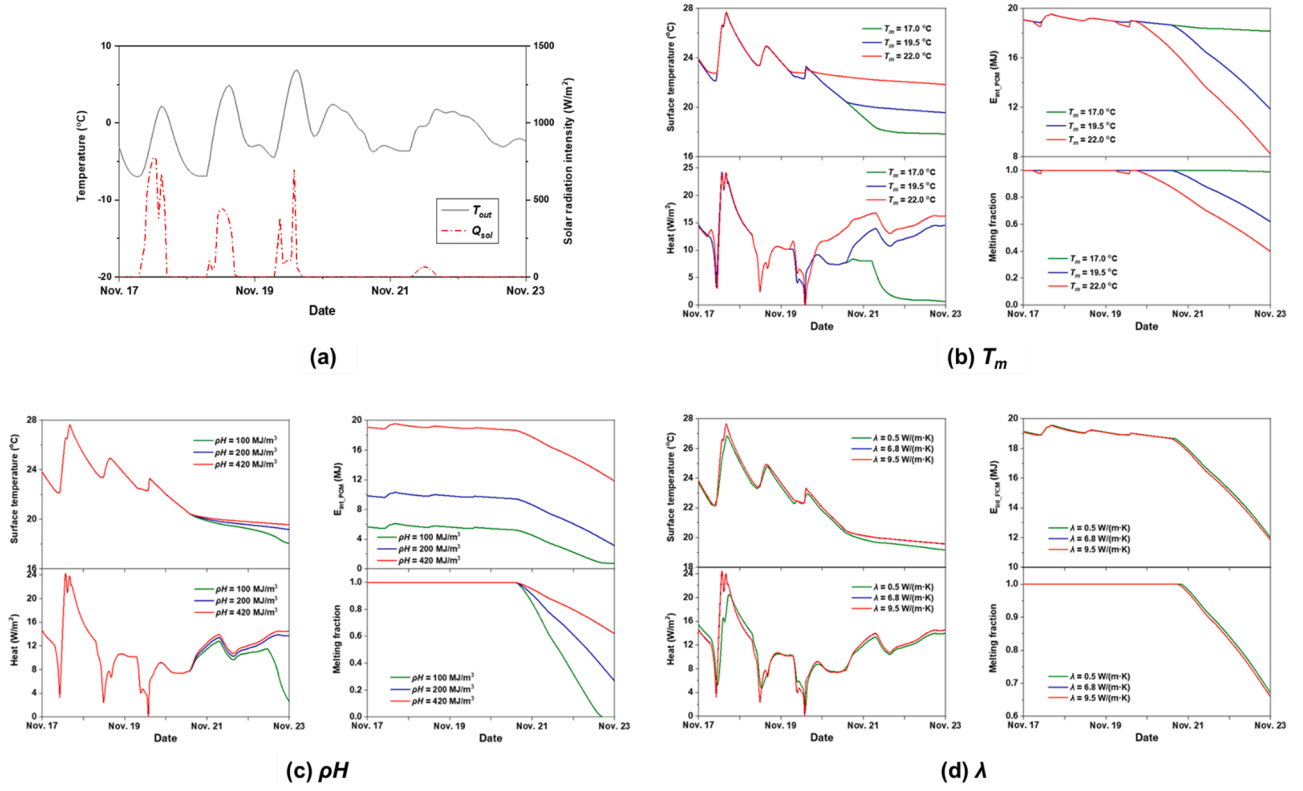


Fig. 13. Thermal performance of the phase change interior wall with different thermophysical properties.

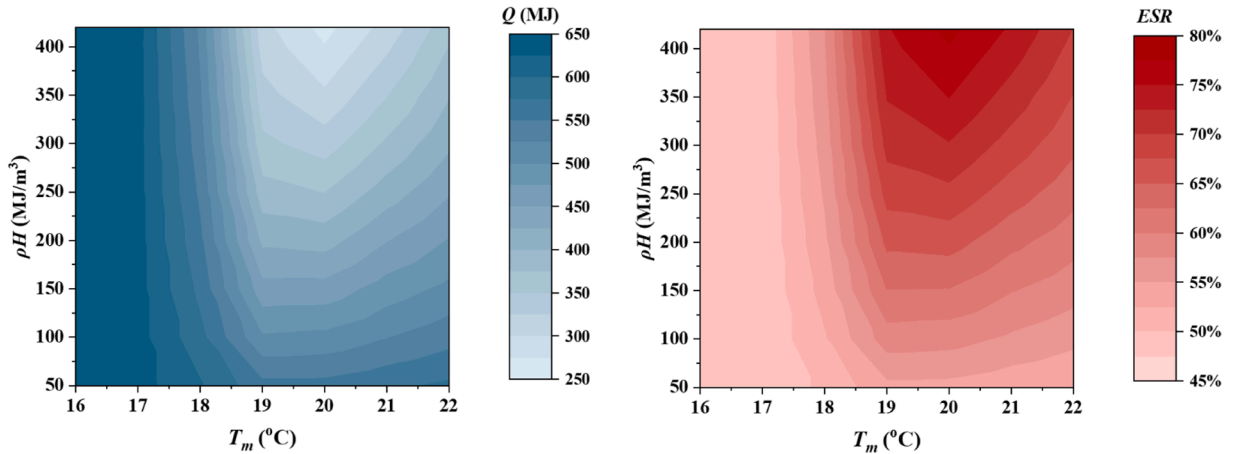


Fig. 14. Influence of ρH and T_m on Q and ESR (Tianjin).

which is consistent with the results of BIHP-PCM without HVAC. It should be noted that the optimal ρH takes the maximum value because of the limitation by the current thermophysical properties of PCM, and it is a reasonable inference that increasing ρH hypothetically would further improve the indoor thermal environment. However, the optimal λ is not its maximum value, because an overhigh λ in the currently feasible range instead leads to premature heat release to the room and is therefore not the optimal option. This result can provide a reference for the research on PCM for solar heating from the angle of material technology.

By comparing Fig. 9 and Fig. 15, it can be seen that different from the results of BIHP-PCM without HVAC, the optimal T_m of BIHP-PCM with HVAC are various but in the range of 19.0 to 20.5 °C, all slightly higher than the lower temperature limit of the thermal comfort zone T_L . It should be noted that the optimal T_m corresponds to the optimal λ , and λ

of the current PCM is much lower, so the T_m under the current λ would be higher than this optimal T_m .

3.3. Evaluation and zoning of application potential

The thermophysical properties of the BIHP-PCM with and without HVAC are optimized and the corresponding optimal heating effects are obtained at all cities, and ESR and $IDTD$ of all non-zero carbon heating cities are calculated. As shown in Fig. 16, ESR and $IDTD$ respectively show a highly linear positive correlation with $RRTD_{HS}$. The fitting formulas are shown as Eqs. (19–22), and the fitting degrees are good with coefficients of determination R^2 higher than 0.98. The fitting formulas help estimate the application potential of BIHP-PCM at cities not selected in the present study. The coefficient of $ESR_{PCM,OPT}$ to $RRTD_{HS}$ is 0.148, which means ESR of the optimized BIHP-PCM can increase by

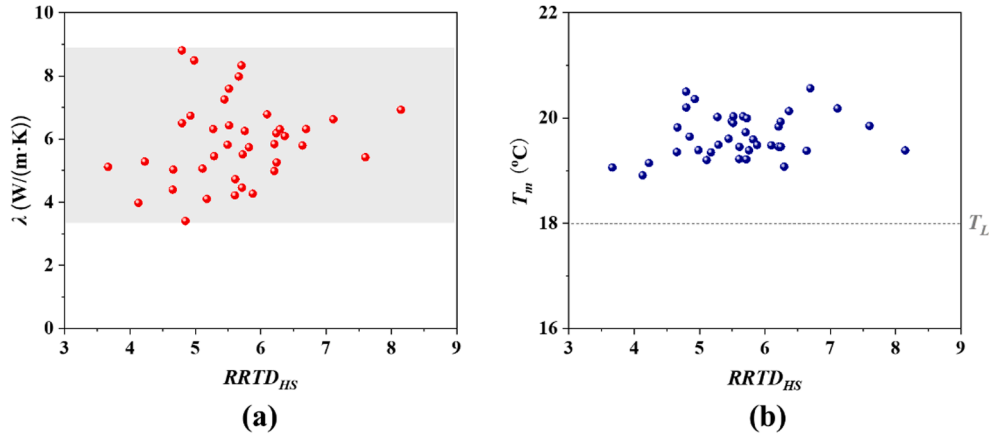


Fig. 15. Optimal thermophysical properties at non-zero carbon heating cities (with HVAC): (a) λ and (b) T_m .

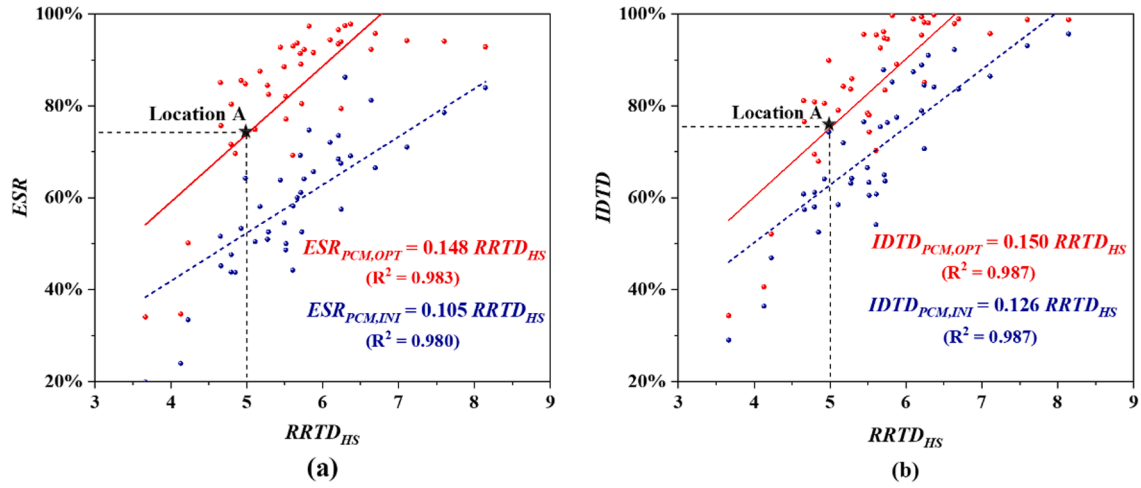


Fig. 16. Linear fitting of (a) ESR and (b) $IDTD$ between $RRTD_{HS}$.

14.8 % for every 1 W/(m²·K) increase of the local $RRTD_{HS}$. Taking location A with the $RRTD_{HS}$ of 5 W/(m²·K) as an example, the maximum ESR by BIHP-PCM with HVAC is predicted to be up to 74.0 %. The same applies to other formulas and coefficients.

$$ESR_{PCM,INI} = 0.105 \cdot RRTD_{HS} \quad (19)$$

$$ESR_{PCM,OPT} = 0.148 \cdot RRTD_{HS} \quad (20)$$

$$IDTD_{PCM,INI} = 0.126 \cdot RRTD_{HS} \quad (21)$$

$$IDTD_{PCM,OPT} = 0.150 \cdot RRTD_{HS} \quad (22)$$

Based on the results of all selected cities, taking ESR and $IDTD$ as the application potential indicators of BIHP-PCM with and without HVAC, respectively, the distribution maps of the application potential are illustrated by ArcGIS and are shown in Fig. 17. Note that only the provinces containing selected cities in this study are shown in the figure. Compared with the heating degree day during the local heating season $HDD18$ and the average solar radiation intensity during the local heating season $Q_{sol,ave}$, the distribution trend of ESR and $IDTD$ are both more consistent to that of $RRTD_{HS}$. It is because $RRTD_{HS}$ reflects the relative intensity of solar radiation at the local outdoor air temperature level, taking into account both the heat gain (solar radiation) and the heat loss (outdoor air temperature) of the solar house.

The BIHP-PCM achieves the best utilization potential at the Qinghai-Tibet Plateau region of western China, where the local solar radiation resources are abundant, with $RRTD_{HS}$ greater than 8.5 W/(m²·K)

generally, and zero-carbon heating is reached. Western Inner Mongolia and northwest Gansu also are regions with good utilization potential of BIHP-PCM, where ESR is greater than 90 % and $IDTD$ is greater than 85 %. On the contrary, the application of BIHP-PCM is relatively weak in northeast China and northern Xinjiang. The solar radiation resources are poor and the outdoor air temperature is low there, with $RRTD_{HS}$ less than 6.0 W/(m²·K). In general, ESR and $IDTD$ reach greater than 30 % at most cities in severe cold and cold regions of China, meaning that BIHP-PCM can significantly reduce heating energy consumption and improve the indoor thermal environment compared with the reference building. Based on the above analysis and the fitting formulas, the application potentiality of BIHP-PCM in severe cold and cold regions of China can be generally categorized into three zones, which are zero-carbon, good, and suitable, as shown in Table 3.

4. Conclusions

In this study, a novel solar heating building integrated with flat gravity heat pipes and phase change materials (BIHP-PCM) is proposed. The thermophysical properties of PCM of the interior wall in BIHP-PCM with HVAC and without HVAC at 61 cities in severe cold and cold regions of China are optimized, and the optimal heating effects are obtained. The relationship between the optimal heating effect of BIHP-PCM and the ratio of solar radiation to air temperature difference is studied. The results show that:

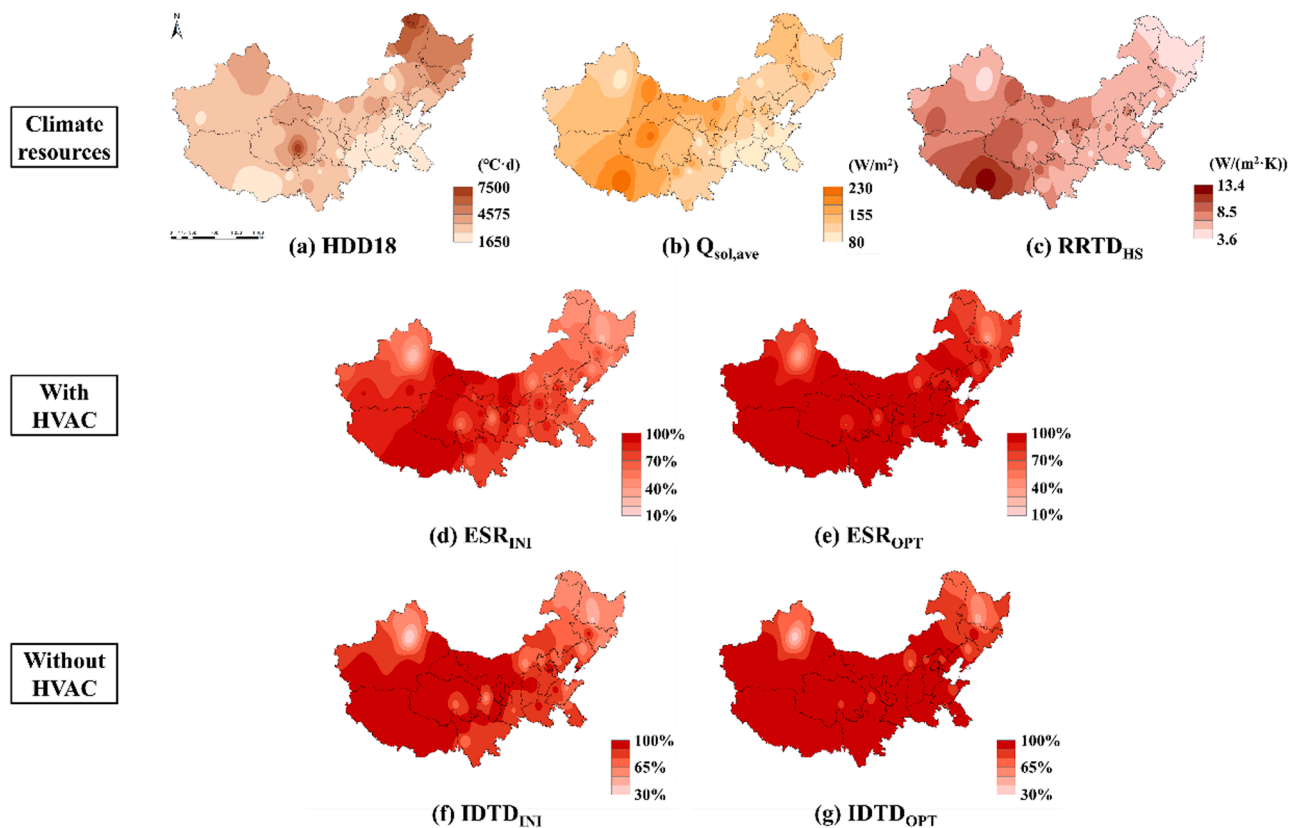


Fig. 17. Spatial distribution of evaluation indexes of climatic resources and utilization potential of BIHP-PCM.

Table 3

Three zones of the application potential of BIHP-PCM.

Zone	$RRTD_{HS}$	ESR	$IDTD$
Zero-carbon	$8.5 \leq RRTD_{HS}$	100 %	100 %
Good	$6.0 \leq RRTD_{HS} < 8.5$	$60 \% \leq ESR < 100 \%$	$70 \% \leq IDTD < 100 \%$
Suitable	$RRTD_{HS} < 6.0$	$ESR < 60 \%$	$IDTD < 70 \%$

- (1) The optimal PCM of the interior wall in BIHP-PCM with the exterior wall of a high thermal resistance should be of high thermal conductivity and maximum volumetric phase change enthalpy, with and without HVAC at all cities.
- (2) For the phase change temperature, the optimal results of BIHP-PCM with HVAC are slightly higher than the lower limit of the thermal comfort temperature zone by about 2 °C at all cities, but the optimal results of BIHP-PCM without HVAC at some cities are lower than the lower limit of the thermal comfort temperature zone, because of the bottom-supporting effect of PCM on room temperature.
- (3) The cumulative heating energy consumption of BIHP-PCM with HVAC is greatly reduced because the HVAC operating time and the average heating load are both reduced.
- (4) The highly linear positive correlation and the coefficients between the ratio of solar radiation to air temperature difference and the heating effect of BIHP-PCM are obtained.
- (5) The energy saving ratio and improvement degree of thermal discomfort of BIHP-PCM at the selected cities are between 30 %

and 100 %, and the application potential of BIHP-PCM in severe cold and cold regions of China can be categorized into three zones: zero-carbon, good, and suitable.

This study shows that the BIHP-PCM, a novel solar heating building, can significantly reduce heating energy consumption and improve the indoor thermal environment in severe cold and cold regions of China, which has important guiding significance to promote clean heating and achieve the dual-carbon goal.

CRediT authorship contribution statement

Fangcheng Kou: Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis. **Nian Zhu:** Software, Data curation. **Xin Wang:** Supervision, Resources, Project administration, Conceptualization. **Yu Zou:** Supervision, Project administration. **Jinhan Mo:** Supervision, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A

The evaluation indexes of climatic resources and indicators of the utilization potential of BIHP-PCM in all selected cities are listed in [Table A1](#).

Table A1

Detailed results in all selected cities.

City	$RRTD_{HS}$	ESR_{INI}	ESR_{OPT}	$IDTD_{INI}$	$IDTD_{OPT}$
Urumqi	3.66	19.9 %	34.0 %	29.0 %	34.3 %
Harbin	4.13	23.9 %	34.7 %	36.4 %	40.6 %
Hailun	4.23	33.4 %	50.2 %	46.9 %	52.1 %
Weifang	4.65	51.6 %	85.1 %	60.8 %	81.1 %
Linjiang	4.66	45.2 %	75.7 %	57.4 %	76.6 %
Mohe	4.79	43.8 %	71.6 %	58.0 %	69.5 %
Fujian	4.80	47.6 %	80.4 %	61.2 %	80.9 %
Shenyang	4.85	43.8 %	69.6 %	52.5 %	67.9 %
Yanji	4.93	53.3 %	85.5 %	64.1 %	80.6 %
Xi'an	4.98	64.2 %	84.8 %	74.3 %	89.9 %
Lanzhou	5.11	50.5 %	74.8 %	58.5 %	79.1 %
Xuzhou	5.18	58.1 %	87.5 %	71.9 %	84.3 %
Zhangbei	5.27	51.0 %	84.5 %	63.2 %	83.7 %
Qiqihar	5.29	52.5 %	82.5 %	64.2 %	85.9 %
Yanzhou	5.44	63.9 %	92.8 %	76.5 %	95.6 %
Jarud Banner	5.49	54.5 %	88.5 %	66.5 %	78.5 %
Hailar	5.51	48.6 %	82.0 %	63.3 %	78.0 %
Karamay	5.52	50.0 %	77.1 %	60.5 %	74.3 %
Hohhot	5.61	44.2 %	69.2 %	54.1 %	70.3 %
Weihai	5.61	58.2 %	93.0 %	60.8 %	95.5 %
Chifeng	5.66	60.0 %	93.7 %	75.4 %	92.6 %
Barkam	5.70	70.7 %	100 %	75.6 %	100 %
Shijiazhuang	5.71	69.2 %	91.4 %	87.9 %	96.2 %
Litang	5.71	61.1 %	89.1 %	65.0 %	94.8 %
Chengde	5.72	52.5 %	80.5 %	63.6 %	83.4 %
Anyang	5.76	64.1 %	92.3 %	76.4 %	94.6 %
Yan'an	5.82	74.7 %	97.4 %	85.2 %	99.7 %
Dandong	5.88	65.7 %	91.6 %	77.5 %	89.1 %
Tianjin	6.10	72.1 %	94.4 %	87.4 %	98.9 %
Yuncheng	6.21	68.4 %	93.5 %	78.9 %	95.4 %
Jinzhou	6.21	73.6 %	96.6 %	88.9 %	99.4 %
Tianshui	6.23	81.3 %	100 %	94.5 %	100 %
Kashgar	6.24	67.5 %	94.0 %	84.5 %	98.2 %
Maduo	6.24	57.5 %	79.4 %	70.7 %	85.1 %
Zhengzhou	6.30	86.2 %	97.5 %	91.0 %	98.1 %
Changchun	6.33	72.4 %	100 %	90.2 %	100 %
Datong	6.37	69.1 %	97.8 %	84.2 %	99.9 %
Dalian	6.47	87.0 %	100 %	97.6 %	100 %
Jinan	6.64	81.2 %	92.3 %	92.3 %	97.9 %
Xilinhot	6.69	66.5 %	95.8 %	83.7 %	98.9 %
Kuqa	7.11	71.0 %	94.2 %	86.5 %	95.7 %
Yulin	7.19	82.3 %	100 %	97.2 %	100 %
Xining	7.20	92.1 %	100 %	99.8 %	100 %
Guyuan	7.20	95.6 %	100 %	100 %	100 %
Taiyuan	7.26	91.1 %	100 %	99.1 %	100 %
Yushu	7.36	97.6 %	100 %	100 %	100 %
Ruoqiang	7.58	92.2 %	100 %	99.5 %	100 %
Zhangye	7.60	78.6 %	94.1 %	93.1 %	98.8 %
Beijing	8.15	84.0 %	92.9 %	95.7 %	98.8 %
Hongyuan	8.15	94.9 %	100 %	100 %	100 %
Yumen	8.34	97.4 %	100 %	100 %	100 %
Urad Rear Banner	8.43	100 %	100 %	100 %	100 %
Ejina Banner	9.05	95.5 %	100 %	100 %	100 %
Yinchuan	9.11	100 %	100 %	100 %	100 %
Hotan	9.19	90.3 %	100 %	99.4 %	100 %
Chamdo	9.29	100 %	100 %	100 %	100 %
Hami	9.33	100 %	100 %	100 %	100 %
Ganzi	9.59	100 %	100 %	100 %	100 %
Golmud	9.65	100 %	100 %	100 %	100 %
Nyingchi	10.97	100 %	100 %	100 %	100 %
Lhasa	13.30	100 %	100 %	100 %	100 %

Data availability

Data will be made available on request.

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